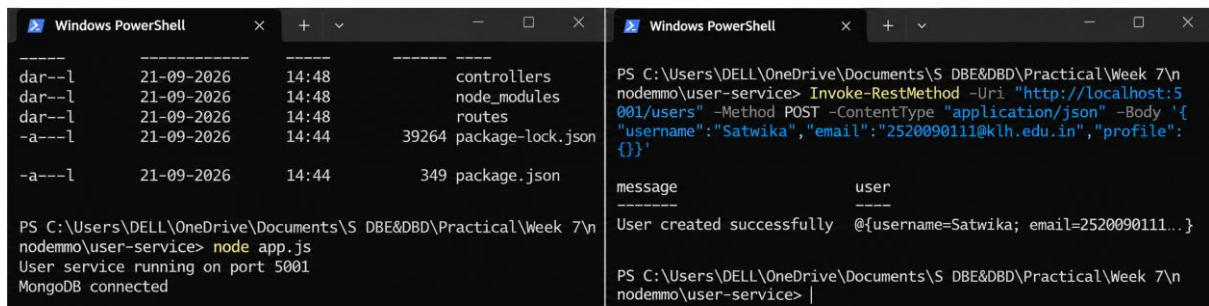


WEEK-7 PRACTICAL — Node.js User Service [G. SATWIKI, 2520090111]

Step – 1 Create a User — POST

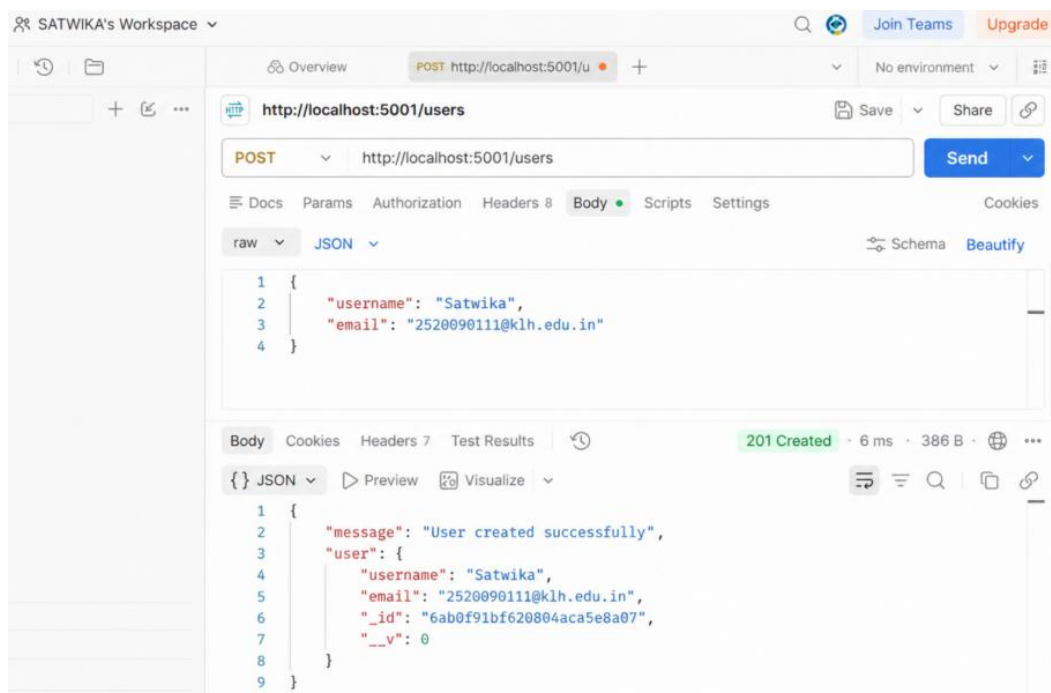


```
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7> dir
dar--l      21-09-2026    14:48      controllers
dar--l      21-09-2026    14:48      node_modules
dar--l      21-09-2026    14:48      routes
-a---l      21-09-2026    14:44    39264 package-lock.json
-a---l      21-09-2026    14:44      349 package.json

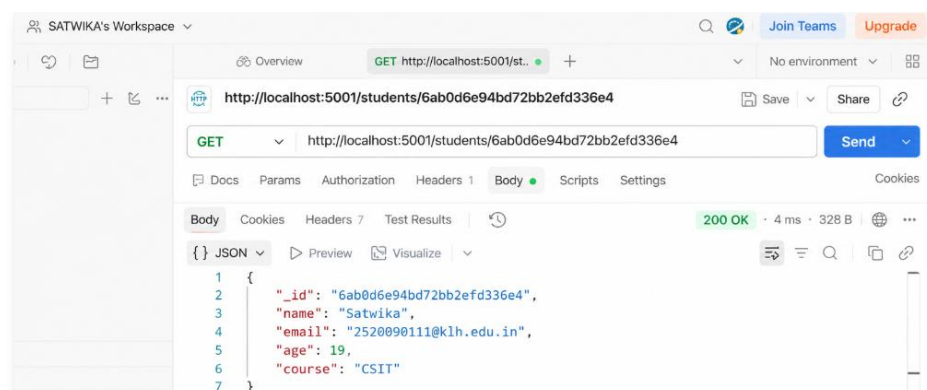
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7> node app.js
nodemmo\user-service> node app.js
User service running on port 5001
MongoDB connected

PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7> Invoke-WebRequest -Uri "http://localhost:5001/users" -Method POST -ContentType "application/json" -Body '{"username": "Satwika", "email": "2520090111@klh.edu.in", "profile": {}}'
message      user
-----
User created successfully @{username=Satwika; email=2520090111...}
```

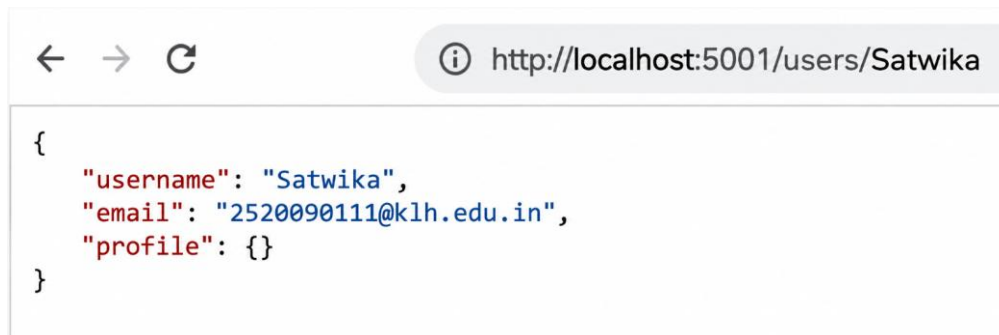
POST in mongosh:



Step 2: Get User — GET



Output on browser by id:

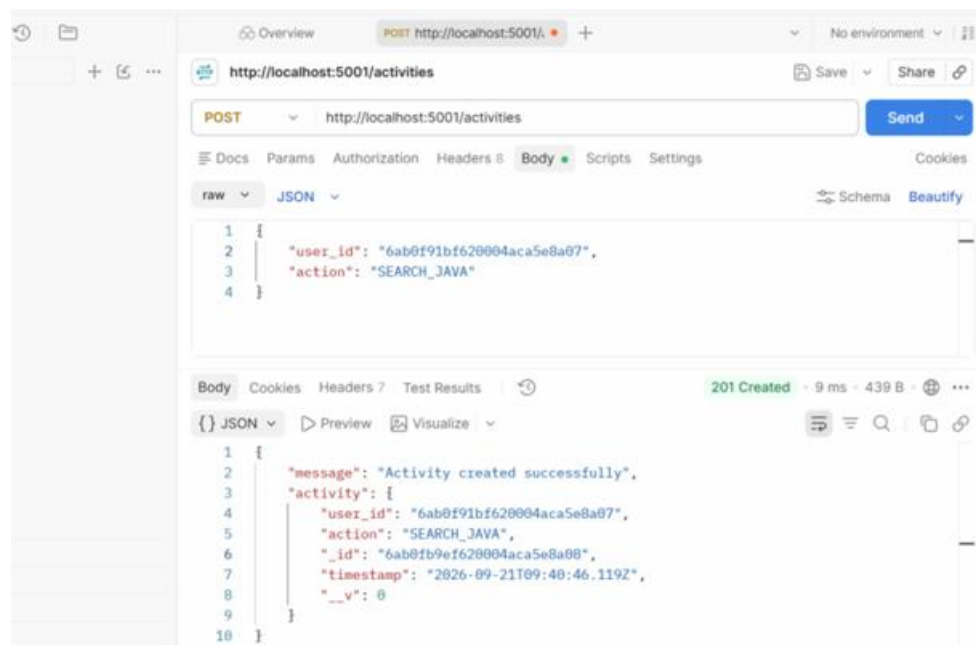


Step 3: Create Activity — POST

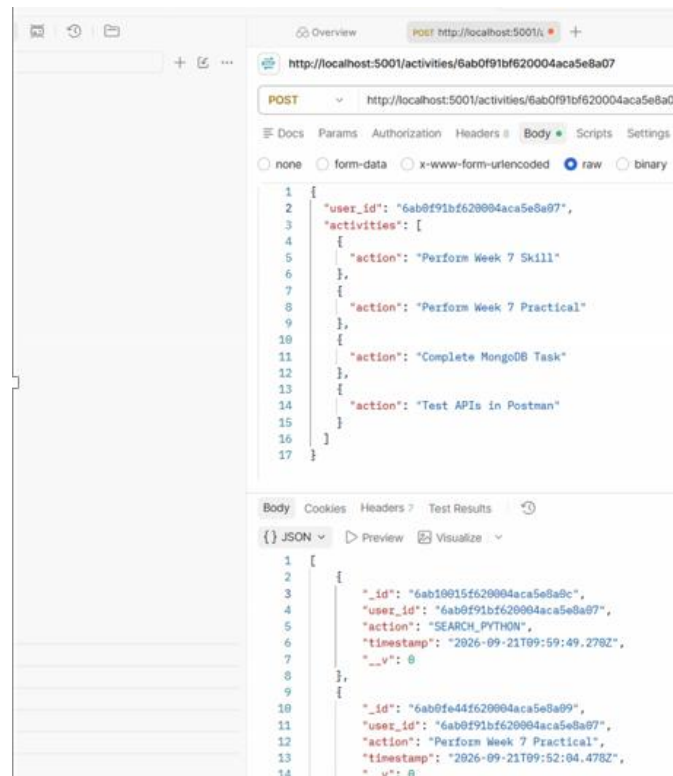
```
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\nodememo\user-service> Invoke-WebRequest -Uri "http://localhost:5001/activities" -Method POST -ContentType "application/json" -Body '{"user_id":"6ab0f91bf620004aca5e8a07","action":"SEARCH_PYTHON"}'

message          activity
-----
Activity created successfully @{user_id=6ab0f91bf620004aca5e...
```

In mongosh:



Step 4: Add More Activities



```

PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\Week 7\nodenemo\user-service> Invoke-WebRequest -Uri "http://localhost:5001/activities/6ab0f91bf620004aca5e8a07" -Method GET

_id      : 6ab10015f620004aca5e8a0c
user_id   : 6ab0f91bf620004aca5e8a07
action    : SEARCH_PYTHON
timestamp : 2026-09-21T09:59:49.270Z
_v        : 0

_id      : 6ab0fe44f620004aca5e8a09
user_id   : 6ab0f91bf620004aca5e8a07
action    : Perform Week 7 Practical
timestamp : 2026-09-21T09:52:04.478Z
_v        : 0

_id      : 6ab0fb9ef620004aca5e8a08
user_id   : 6ab0f91bf620004aca5e8a07
action    : SEARCH_JAVA
timestamp : 2026-09-21T09:40:46.119Z
_v        : 0

```

Output on browser:

localhost:5001/activities/6ab0f91bf620004aca5e8a07

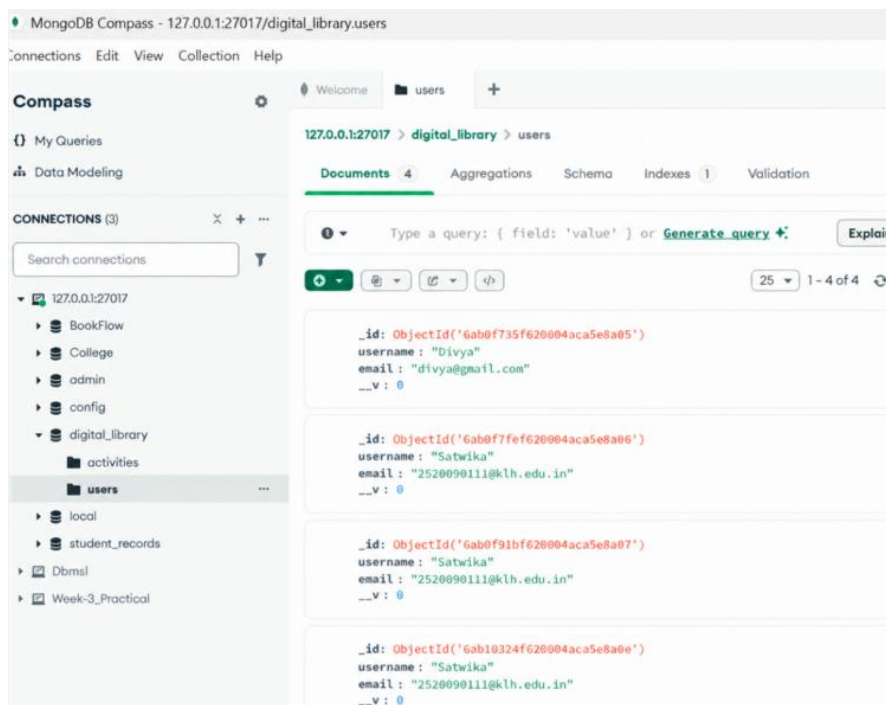
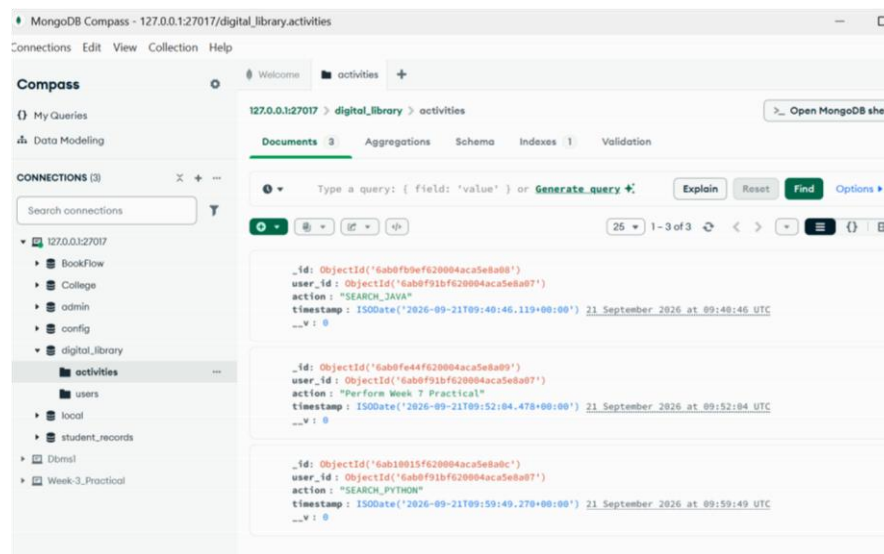
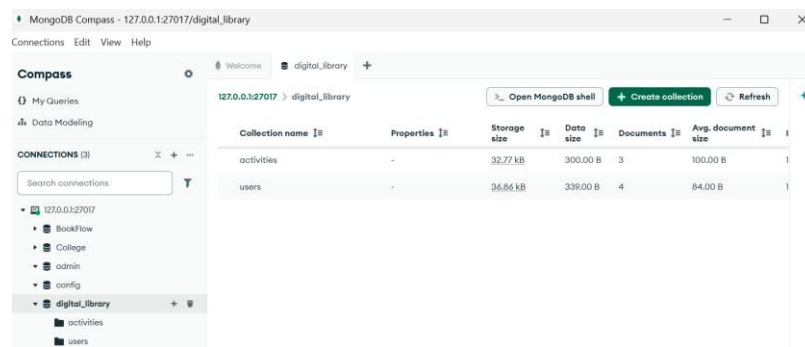
Pretty print ☒

```

[
  {
    "_id": "6ab10015f620004aca5e8a0c",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "SEARCH_PYTHON",
    "timestamp": "2026-09-21T09:59:49.270Z",
    "_v": 0
  },
  {
    "_id": "6ab0fe44f620004aca5e8a09",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "Perform Week 7 Practical",
    "timestamp": "2026-09-21T09:52:04.478Z",
    "_v": 0
  },
  {
    "_id": "6ab0fb9ef620004aca5e8a08",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "SEARCH_JAVA",
    "timestamp": "2026-09-21T09:40:46.119Z",
    "_v": 0
  }
]

```

MongoDB connection:



Folder Structure:

Week 7

```
|__nodememo
  |_user-service
    |
    ├── controllers
    |   ├── activityController.js
    |   └── userController.js
    |
    ├── routes
    |   ├── activityRoutes.js
    |   └── userRoutes.js
    |
    ├── app.js
    ├── db.js
    ├── package.json
    └── .env
```

db.js

```
const mongoose = require("mongoose");
const MONGO_URI =
process.env.MONGO_URI ||
  "mongodb://127.0.0.1:27017/digital_library";
mongoose.connect(MONGO_URI)
  .then(() => console.log("MongoDB connected"))
  .catch(err => console.error("MongoDB connection error:", err));

const userSchema = new mongoose.Schema({
  username: {          type: String,
    required: true
  },    email: {
    type: String,
    required: true
  },    profile: {          type:
mongoose.Schema.Types.Mixed,
    default: {}
  }
}); const activitySchema = new
mongoose.Schema({    user_id: {          type:
mongoose.Schema.Types.ObjectId,
    required: true,          ref: "User"
  },    action: {
    type: String,
    required: true
  },    timestamp: {
    type: Date,
    default: Date.now
  }
}); const User = mongoose.model("User",
userSchema);
const Activity = mongoose.model("Activity", activitySchema);

module.exports = { User, Activity };
```

controllers/userController.js

```
const { User } = require("../db");
const createUser = async (req, res) => {    try {
const { username, email, profile } = req.body;
    const user = await User.create({
  username,          email,
  profile
});
    res.status(201).json(user);
  } catch (err) {
    res.status(500).json({ message: err.message });
  }
};
```

```

    });
    res.status(201).json({
message: "User created successfully",
user
    });
    } catch (error) {
res.status(500).json({
message: "Error
creating user",
error: error.message
    });
    }
};
const getUser = async (req, res) => {
    try {
const user = await User.findById(req.params.id);
        if (!user)
            return res.status(404).json({
message: "User not found"
            });
        res.json(user);
    } catch
(error) {
        res.status(500).json({
message: "Error getting user",
error: error.message
        });
    }
};
module.exports = { createUser, getUser
};

```

controllers/activityController.js

```

const mongoose = require("mongoose"); const
{ Activity } = require("../db");
const createActivity = async (req, res) => {
try {
    const { user_id, action } =
req.body;
        if (!mongoose.Types.ObjectId.isValid(user_id))
return res.status(400).json({
message:
"Invalid user_id"
        });
        const activity = await Activity.create({
user_id,
action
        });
        res.status(201).json({
message: "Activity created successfully",
activity
        });
    } catch (error) {
res.status(500).json({
message: "Error
creating activity",
error: error.message
    });
}

```

```

    }
};

const getActivities = async (req, res) => {
  try {
    const { user_id } =
req.params;

    if (!mongoose.Types.ObjectId.isValid(user_id))
return res.status(400).json({
message:
"Invalid user_id"
});

    const activities = await Activity
.find({ user_id })
.sort({ timestamp: -1 });

    res.json(activities);
  } catch
(error) {
    res.status(500).json({
message: "Error getting activities",

```

```

        error: error.message
    });
}
}; module.exports = { createActivity, getActivities
};

```

routes/userRoutes.js

```

const express = require("express");
const router = express.Router();
const {
  createUser,
  getUser
} = require("../controllers/userController");
router.post("/users", createUser);
router.get("/users/:id", getUser);

```

```

module.exports = router;

```

routes/activityRoutes.js

```

const express = require("express");
const router = express.Router();
const {
  createActivity,
  getActivities
} = require("../controllers/activityController");

router.post("/activities", createActivity);
router.get("/activities/:user_id", getActivities);

```

```

module.exports = router;

```

```

app.js const express =
require("express");
const app = express();

```

```

require("./db");
app.use(express.json());
const activityRoutes =
require("./routes/activit
yRoutes"); const
userRoutes =
require("./routes/userRou
tes");
app.use("/", activityRoutes);
app.use("/", userRoutes);

```

```
app.listen(5001, () => {    console.log("User service  
running on port 5001");  
});
```